

Lagrangian Derivation for an n -Link Spring-Jointed Panel Array with Inflatable Kink Actuation

Arael Anaya
Senior Design

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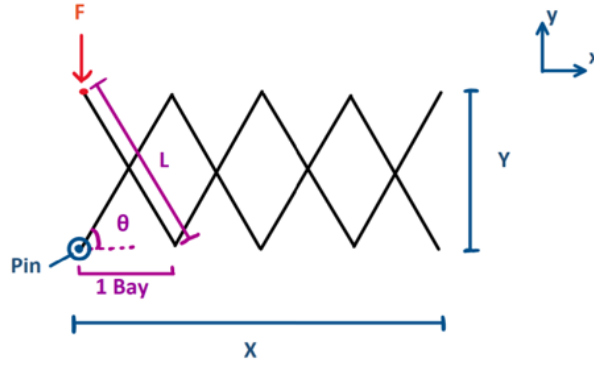


Figure 1: Scissor mechanism geometry and coordinate definition

Notation: $s\theta \equiv \sin \theta$, $c\theta \equiv \cos \theta$, and $\hat{n}_1, \hat{n}_2$ are the body-frame unit vectors. The single generalized coordinate is $q = \theta$. V denotes hinge-spring potential energy; V_g denotes enclosed gas volume.

1 Position Kinematics

Center of mass of the first rod:

$$\vec{r}_1 = \frac{1}{2}Lc\theta \hat{n}_1 + \frac{1}{2}Ls\theta \hat{n}_2 \quad (1)$$

Center of mass of the second rod (offset by one full link):

$$\vec{r}_2 = L \cos \theta + \frac{1}{2}Lc\theta \quad (2)$$

Generalizing to the i -th rod:

$$\vec{r}_i = \left[(i-1)Lc\theta + \frac{1}{2}Lc\theta \right] \hat{n}_1 + \frac{1}{2}Ls\theta \hat{n}_2 \quad (3)$$

2 Velocity Kinematics

Differentiating in time:

$$\vec{v}_1 = \left[-\frac{1}{2}Ls\theta\dot{\theta} \right] \hat{n}_1 + \left[\frac{1}{2}Lc\theta\dot{\theta} \right] \hat{n}_2 \quad (4)$$

$$\vec{v}_2 = -Ls\theta\dot{\theta} - \frac{1}{2}Ls\theta\dot{\theta} = -\frac{3}{2}Ls\theta\dot{\theta} \quad (5)$$

$$\vec{v}_i = \left(\frac{1}{2} - i \right) Ls\theta\dot{\theta} \hat{n}_1 \quad (6)$$

Check of the index coefficient in (6):

$$\frac{1}{2} - 1 = -\frac{1}{2}, \quad \frac{1}{2} - 2 = -\frac{3}{2} \quad (7)$$

which reproduces $\vec{v}_1$ and $\vec{v}_2$.

Index identity

The squared coefficient that appears in the kinetic energy expands as

$$\left(\frac{1}{2} - i \right)^2 = \left(\frac{1}{2} \right)^2 - i + i^2 = i(i-1) + \left(\frac{1}{2} \right)^2 = i(i-1) + \frac{1}{4} \quad (8)$$

3 Kinetic Energy

The total kinetic energy splits by component group:

$$T = T_{\text{rods}} + T_{\text{supports}} + T_{\text{panels}} \quad (9)$$

Summing translational and rotational contributions over the n links:

$$T = \sum_{i=1}^n \left[\frac{1}{2}M_j \left(\frac{1}{2} - i \right)^2 L^2 s^2 \theta \dot{\theta}^2 + \frac{1}{2}I_j \dot{\theta}^2 \right] \quad (10)$$

Substituting the identity (8) into (10):

$$T = \sum_{i=1}^n \left[\frac{1}{2}M_j i(i-1) L^2 s^2 \theta \dot{\theta}^2 + \frac{1}{8}M_j L^2 s^2 \theta \dot{\theta}^2 + \frac{1}{2}I_j \dot{\theta}^2 \right] \quad (11)$$

4 Potential Energy — Hinge Springs

Energy stored in the N spring joints:

$$V = \frac{1}{2}k\theta^2 \cdot N \quad (12)$$

5 Pressure Generalized Force

Virtual work

The gas exerts a traction $P\hat{n}$ on the wetted membrane, so under a virtual displacement $\delta\vec{r}$ of that surface,

$$\delta W_p = \Delta P \oint_S \hat{n} \cdot \delta\vec{r} dA = \Delta P \delta V_g, \quad \Delta P = P_{int} - P_{ext} \quad (13)$$

since the surface integral is the first variation of enclosed volume. Specializing to $\delta V_g = (\partial V_g / \partial \theta) \delta \theta$:

$$Q_p = \Delta P \frac{\partial V_g}{\partial \theta} \quad (14)$$

On orbit $P_{ext} = 0$ so $\Delta P = P_{int}$.

Kink deviation angle

At a kink the tube arrives along $(c\theta, s\theta)$ and departs along $(c\theta, -s\theta)$, so the deviation from straight is

$$\beta = 2\theta, \quad \frac{d\beta}{d\theta} = 2 \quad (15)$$

At $\theta = \pi/2$: $\beta = \pi$ (folded flat, stowed, max deficit). At $\theta = \theta_{min}$: $\beta = 2\theta_{min}$ (kinks nearly open).

Volume function

Writing volume as a straight-tube baseline less a deficit at each of N_k kinks:

$$V_g(\theta) = V_0 - N_k \Delta V(\beta) = V_0 - N_k \Delta V(2\theta) \quad (16)$$

with $V_0 = \pi R^2 s_{tube}$ (R = inflated tube radius) and $\Delta V(0) = 0$, increasing in β .

Kink moment

Substituting (16) into (14) via the chain rule:

$$Q_p = \Delta P \frac{\partial V_g}{\partial \theta} = -2N_k \Delta P \left. \frac{d\Delta V}{d\beta} \right|_{\beta=2\theta} \quad (17)$$

Defining the kink moment

$$M_k(\beta, \Delta P) \equiv \Delta P \frac{\partial \Delta V}{\partial \beta} \quad \implies \quad \boxed{Q_p = -2N_k M_k(2\theta, \Delta P)} \quad (18)$$

$M_k > 0$, so $Q_p < 0$: pressure drives θ down from $\pi/2$ toward θ_{min} , same sense as the hinge springs. M_k appears in the same slot as $k(\theta - \theta_0)$. Essentially the pressurized tube is a nonlinear torsional spring in parallel with the hinge springs.

Closure forms

Dimensional scaling gives $M_k = a \Delta P R^3 f'(\beta)$. Two candidate closures:

$$(C1) \text{ constant fold moment: } M_k = a \Delta P R^3 \quad (19)$$

$$(C2) \text{ smooth vanishing fold: } M_k = a \Delta P R^3 \sin 2\theta \quad (20)$$

6 Lagrangian and Equation of Motion

$$\mathcal{L} = T - V, \quad H = T + V \quad (21)$$

The forced Euler–Lagrange equation with $q = \theta$:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = Q_p + Q_F \quad (22)$$

Momentum term

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = M_j i(i-1) L^2 s^2 \theta \dot{\theta} + \left[I_j + \frac{1}{2} M_j L^2 \right] \dot{\theta} \quad (23)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = M_j i(i-1) L^2 s^2 \theta \ddot{\theta} + 2 M_j i(i-1) L^2 s \theta c \theta \dot{\theta}^2 + \left[I_j + \frac{1}{2} M_j L^2 \right] \ddot{\theta} \quad (24)$$

Coordinate term

$$\frac{\partial \mathcal{L}}{\partial \theta} = M_j i(i-1) L^2 \dot{\theta}^2 s \theta c \theta - k \theta \quad (25)$$

Assembled equation

Substituting (24), (25), and Q_p from (18) into (22), with the two $s \theta c \theta \dot{\theta}^2$ terms combining ($2 - 1 = 1$), and moving Q_p to the LHS alongside the spring term (it is likewise a function of θ alone at fixed ΔP):

$$\boxed{\left[M_j i(i-1) L^2 s^2 \theta + I_j + \frac{1}{2} M_j L^2 \right] \ddot{\theta} + M_j i(i-1) L^2 s \theta c \theta \dot{\theta}^2 + k \theta + 2 N_k M_k(2\theta, \Delta P) = Q_F} \quad (26)$$

7 PHASE 1

From $\theta = \pi/2$ till the system no longer experiences the moment singularity due to kinking, we will rely solely on the hinged springs to generate enough force in the system. As such we have a

simplified version of the Lagrange with no pressure force ($Q_p = Q_F = 0$):

$$\underbrace{\left[M_j i(i-1)L^2 s^2 \theta + I_j + \frac{1}{2} M_j L^2 \right]}_{I_{\text{eff}}(\theta)} \ddot{\theta} + M_j i(i-1)L^2 s \theta c \dot{\theta}^2 + N_s k \theta = 0 \quad (27)$$

This diff eq can be solved using **conservation of energy (first integral of motion)**, since the system is scleronomic and unforced here: $H = T + V$ is conserved.

$$\frac{1}{2} I_{\text{eff}}(\theta) \dot{\theta}^2 + \frac{1}{2} N_s k \theta^2 = E \quad (28)$$

Assuming a desired duration of t , starting from rest at $\theta(0) = \pi/2$, so $E = \frac{1}{2} N_s k (\pi/2)^2$:

$$\dot{\theta} = -\sqrt{\frac{N_s k [(\pi/2)^2 - \theta^2]}{I_{\text{eff}}(\theta)}} \quad (29)$$

Integrating from stowage to the Phase 1 endpoint θ_1 :

$$t = \frac{1}{\sqrt{N_s k}} \int_{\theta_1}^{\pi/2} \sqrt{\frac{I_{\text{eff}}(\theta)}{(\pi/2)^2 - \theta^2}} d\theta \quad (30)$$

The required spring constant (based on all known mass parameters) is:

$$k = \frac{1}{N_s t^2} \left[\int_{\theta_1}^{\pi/2} \sqrt{\frac{I_{\text{eff}}(\theta)}{(\pi/2)^2 - \theta^2}} d\theta \right]^2 \quad (31)$$

8 PHASE 2

Past θ_1 , kinking becomes reliable and the pressure generalized force Q_p enters.

$$I_{\text{eff}}(\theta) \ddot{\theta} + M_j i(i-1)L^2 s \theta c \dot{\theta}^2 + N_s k \theta + 2N_k M_{k0} = 0 \quad (32)$$

where $M_{k0} \equiv M_k|_{C1}$ is the unknown constant kink moment to be solved for.

Because M_{k0} is constant, Q_p is still conservative: define an actuator potential referenced to the Phase 1/2 boundary,

$$U_p(\theta) = 2N_k M_{k0} (\theta - \theta_1), \quad U_p(\theta_1) = 0 \quad (33)$$

Boundary matching

Continuity with Phase 1 fixes the initial condition of Phase 2 at θ_1 :

$$\dot{\theta}_1 = -\sqrt{\frac{N_s k [(\pi/2)^2 - \theta_1^2]}{I_{\text{eff}}(\theta_1)}} \quad (34)$$

Energy conservation

$$\frac{1}{2} I_{\text{eff}}(\theta) \dot{\theta}^2 + \frac{1}{2} N_s k \theta^2 + 2 N_k M_{k0} (\theta - \theta_1) = \frac{1}{2} I_{\text{eff}}(\theta_1) \dot{\theta}_1^2 + \frac{1}{2} N_s k \theta_1^2 \quad (35)$$

Solving for $\dot{\theta}$:

$$\dot{\theta} = -\sqrt{\frac{I_{\text{eff}}(\theta_1) \dot{\theta}_1^2 + N_s k (\theta_1^2 - \theta^2) - 4 N_k M_{k0} (\theta - \theta_1)}{I_{\text{eff}}(\theta)}} \quad (36)$$

Integrating to the full-deployment angle $\theta_{\min}$, the desired Phase 2 duration t_2 implicitly determines the required kink moment:

$$t_2 = \int_{\theta_{\min}}^{\theta_1} \sqrt{\frac{I_{\text{eff}}(\theta)}{I_{\text{eff}}(\theta_1) \dot{\theta}_1^2 + N_s k (\theta_1^2 - \theta^2) - 4 N_k M_{k0} (\theta - \theta_1)}} d\theta \quad (37)$$

Unlike Phase 1, this does not invert cleanly for M_{k0} (it sits inside the integrand as well as scaling the whole bracket), so it must be solved numerically, e.g. root-find M_{k0} so this integral equals the target t_2 .